

Emotion-Controlled Text Rewriting with Transformers: Balancing Emotional Expressiveness and Semantic Preservation

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Abstract

Emotion-controlled text rewriting is a challenging task in natural language processing because increasing emotional expressiveness often changes the original meaning of a sentence. In this project, we study the trade-off between emotional control and semantic preservation in transformer-based text rewriting. We fine-tune a pretrained BART model to rewrite neutral sentences into text expressing a target emotion, specifically joy, anger, or sadness, while preserving the core meaning of the original input. To evaluate emotional accuracy, we fine-tune a BERT-based classifier on a filtered 4-class version of the GoEmotions dataset. To evaluate semantic preservation, we compute cosine similarity between Sentence-BERT embeddings of the original and generated sentences. Our results show that the classifier achieves strong validation performance, with about 91% accuracy and a macro-F1 score around 0.78. Qualitative and quantitative results suggest that the model can generate emotionally distinguishable outputs, but balancing emotional expressiveness and semantic preservation remains the central challenge.

1 Problem Statement and Motivation

Controllable text generation, specifically emotion-based expression, is a core part of natural language processing as LLMs become more advanced with applications in empathetic chat-bots, creative writing, and content personalization. A model that can express the emotional tone of a sentence can unintentionally shift its meaning, creating a trade-off between expressiveness and semantic validity. Our reasons for wanting to experiment with altering the emotional tone of a sentence in LLMs is to enhance the user experience on the front end. Not everyone interprets tone in the same way, and enhancing the emotional delivery can bridge communication gaps when a user is not understanding

information given in a neutral tone. Mastering the emotional tone a sentence conveys is just as important as the content or information that the sentence is trying to convey. This is the fundamental issue of focusing on semantic and emotional scoring. Our project addresses the task of emotion-controlled text rewriting: given a neutral input sentence and target emotion, generate a rewritten sentence that expresses the desired emotion while preserving the original meaning. We aim to analyze the tradeoff and evaluate using a transformer-based model.

2 Related Work

This project is based on previous works done in the field of Emotion Style Transfer (EST) and controllable text generation, with its main objective of modifying specific attributes of a sentence while keeping its core content intact. Our work builds upon the sequence-to-sequence capabilities of the BART architecture [Debnath et al. \(2024\)](#) with the GoEmotions ([Demszky et al., 2020](#)) dataset: a large-scale, human-annotated dataset of Reddit comments with 27 emotion categories, which has been used for emotional classification and generation tasks. Existing research, such as work by [Troiano et al. \(2020\)](#), has highlighted a persistent "tension" in emotional rewriting where emotional intensity and semantic fidelity have an inverse relationship.

Early approaches to emotion style transfer worked without paired training data and without inputs that have a neutral sentence paired with its emotional rewrite. Instead, it tried to split a sentence into two separate parts which was one part capturing *what* is being said (content) and another capturing *how* it is said (style). [Shen et al. \(2017\)](#) trained two networks where one would rewrite text with a target style while the other one judged whether the style change was convincing. [Hu](#)

et al. (2017) took a similar idea but used a Variational Autoencoder (VAE), a model that compresses a sentence into a compact representation and then reconstructs it. While these approaches were pioneering, they tend to produce outputs that drift from the original meaning or read unnaturally. This is because keeping the content and style cleanly separated in practice is very difficult to guarantee. This shortcoming motivates the use of supervised sequence-to-sequence models in our work, which instead learn directly from matched pairs of neutral and emotional sentences, making the rewriting task more explicit and the outputs more reliable.

A related area of research is controllable text generation, where the goal is to guide a language model to produce output with specific constraints.. Keskar et al. (2019) proposed CTRL, a large language model that was trained with special control words prepended to the input, signaling to the model what kind of text it should generate. For example, prepending a word like *News* would cause the model to write in a news article style. This is the direct inspiration behind the emotion tags (e.g., [JOY], [ANGER]) used in our approach, where the tag tells the model which emotion the rewritten sentence should express.

3 Approach

Our approach consists of two components: a BART-based emotion rewriting model and an automatic evaluation pipeline.

We fine-tune `facebook/bart-base` for emotion-controlled text rewriting. BART (Lewis et al., 2019) is a transformer-based sequence-to-sequence model with a bidirectional encoder and an autoregressive decoder, which makes it suitable for rewriting tasks where the output should preserve the core content of the input while changing its style or tone. In our task, the model takes a neutral sentence and a target emotion as input, and generates a rewritten sentence that expresses the desired emotion.

The input to the BART model is formatted as:

```
emotion: <target_emotion> | text: <
  neutral_sentence>
```

This prefix-based format is inspired by controllable text generation methods such as CTRL, where control tags are used to guide the style of generated text. In our case, the control tag specifies the target emotion category.

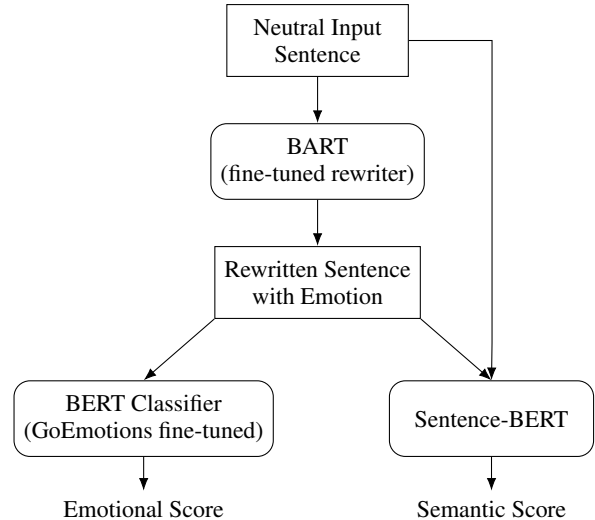


Figure 1: Emotion rewriting pipeline with semantic and emotional evaluation.

Because the GoEmotions dataset does not provide gold rewrite pairs, we construct a pseudo-parallel rewriting dataset. Specifically, we pair neutral sentences with semantically similar emotion-labeled sentences using Sentence-BERT embeddings and cosine similarity. These pseudo-pairs are then used to fine-tune the BART rewriting model.

For automatic evaluation, we use two complementary scoring components. First, we fine-tune a BERT-based classifier (`bert-base-uncased`) on a filtered 4-class version of GoEmotions containing *neutral*, *joy*, *anger*, and *sadness*. This classifier is used to predict the emotion label of each generated sentence and provide an emotional score. Second, we use Sentence-BERT to compute cosine similarity between the original neutral sentence and the generated sentence as a semantic score.

Together, these two components form an evaluation pipeline that measures both emotional expressiveness and semantic preservation. This is important because the central challenge of emotion-controlled rewriting is balancing these two objectives: stronger emotional expression can improve style control, but it can also increase semantic drift from the original input.

4 Experimental Setup

4.1 Data

We use a filtered subset of the GoEmotions dataset containing four categories: *neutral*, *joy*, *anger*, and *sadness*. After filtering, the dataset contains

approximately 18,000 training examples, 2,300 validation examples, and 2,300 test examples. The original data consists of Reddit comments, which provides short informal text suitable for emotion classification and controlled rewriting experiments.

For the rewriting model, GoEmotions does not provide true parallel pairs of neutral sentences and their emotional rewrites. Therefore, we built a pseudo-parallel dataset by pairing neutral inputs with semantically similar sentences from the target emotion classes using Sentence-BERT similarity.

4.2 BERT-based Emotion Classifier

We fine-tune `bert-base-uncased` as a 4-class emotion classifier on the filtered GoEmotions dataset. The classifier is used as an automatic evaluator for generated sentences. Its output includes both a predicted emotion label and a confidence score.

The main hyperparameters are:

- Learning rate: 2×10^{-5}
- Batch size: 16
- Training epochs: 3
- Optimizer: AdamW

We also experimented with class-weighted loss to address class imbalance, but it did not improve validation macro-F1 in our final runs, so the final classifier uses the standard training objective.

4.3 BART-based Rewriter

We fine-tune `bart-base` for sequence-to-sequence emotion rewriting. The model is trained with cross-entropy loss using the pseudo-parallel rewriting dataset.

The main hyperparameters are:

- Learning rate: 3×10^{-5}
- Batch size: 4
- Training epochs: 5
- Loss function: Cross-Entropy Loss

4.4 Evaluation

We use two automatic evaluation dimensions:

- **Emotional Score:** the predicted emotion label and confidence score from the fine-tuned BERT classifier.

- **Semantic Score:** cosine similarity between Sentence-BERT embeddings of the original neutral sentence and the generated sentence.

This evaluation pipeline allows us to analyze the trade-off between emotional expressiveness and semantic preservation across different target emotions.

4.5 Implementation

Models will be implemented using the HuggingFace Transformers library with PyTorch. BART will be fine-tuned with cross-entropy loss, batch size 16–32, and learning rate 3×10^{-5} for 3–5 epochs. BERT classifier will be fine-tuned similarly for emotion classification.

5 Timeline

Week	Jeff Chiang	Nicholas Susanto	Yvan Zhang
1	Start classifier baseline	Prepare BART formatting and tokenizer	Set up Sentence-BERT pipeline
2	Train BERT classifier	Debug training pipeline	Prepare evaluation scripts
3	Train rewriting model	Implement cosine similarity scoring	Build automatic evaluation pipeline
4	Milestone experiments and debugging	Milestone experiments and debugging	Milestone experiments and debugging
5	Correlation plots and analysis	Correlation plots and analysis	Correlation plots and analysis
6	Write Method section	Write Evaluation section	Write Analysis and Results

6 Results

6.1 Emotion Classifier Performance

Our final BERT-based emotion classifier achieved strong validation performance on the 4-class GoEmotions subset. The model reached approximately 91% validation accuracy and a macro-F1 score around 0.78. The difference between accuracy and macro-F1 suggests that class imbalance remains a challenge, especially because neutral samples are more common than the other emotion categories.

These results indicate that the classifier is reliable enough to serve as the emotional evaluation component in our automatic evaluation pipeline.

6.2 Automatic Evaluation Summary

Using the fine-tuned classifier and Sentence-BERT semantic scoring, we evaluated rewritten

outputs across the three target emotions: joy, anger, and sadness. The results show that the rewriter can generate outputs with recognizable emotional differences, but performance varies across emotions.

Metric	Score
Average Emotion Classifier Confidence	0.765
Average Semantic Similarity	0.738
Emotion Match Accuracy	0.721

Table 1: Overall performance of the emotion-controlled rewriting model.

These results indicate that the model is generally successful in producing emotionally aligned outputs while maintaining a reasonable level of semantic consistency.

Emotion	Conf.	Sem. Score	Accuracy
Joy	0.89	0.75	1.0
Anger	0.65	0.76	0.2683
Sadness	0.75	0.71	0.9

Table 2: Performance by target emotion.

Joy tends to receive the highest emotional confidence, while anger and sadness show a different balance between emotional strength and semantic preservation. This suggests that positive emotions are easier for the model to predict.

Overall, the automatic evaluation supports the main hypothesis of the project: increasing emotional distinctiveness can make it harder to preserve the original meaning perfectly, so the core challenge is to balance both objectives rather than maximize only one.

6.3 Comparison with Pseudo-Pair Baseline

We also compared our manually supervised rewriting approach against a baseline trained on pseudo sentence pairs. The pseudo-pair setup had neutral sentences paired with the target emotion labels but without the corresponding human written rewrites. This provides a weaker direction for the model as it is not shown how to transform input while preserving meaning and only relies on the emotion classifier to signal whether the target sentiment was achieved.

As a result, the pseudo-pair baseline produces noticeably lower quality outputs. These generations exhibit weaker emotional expression as well as semantic similarity to the original sen-

tence. In contrast to training on pseudo sentence pairs, the manually created dataset enables the model to learn to be more fluent and dynamic with stronger emotion classification confidence and semantic similarity metrics.

The following examples show outputs generated by the pseudo-pair baseline model:

Input: emotion: joy | text: The package arrived
Output: I’m so glad you got it!

Input: emotion: anger | text: The package arrived
Output: I’m so glad you didn’t get it.

Input: emotion: sadness | text: The package arrived
Output: I can’t stand it.

And contrasting to training on the manually created dataset:

Input: emotion: joy | text: It is raining in Vancouver
Output: It is raining in Vancouver, and it is such a nice change of scenery.

Input: emotion: anger | text: It is raining in Vancouver
Output: It is raining in Vancouver right now, and it is driving me crazy.

Input: emotion: sadness | text: It is raining in Vancouver
Output: It is raining in Vancouver right now, and it just makes the place feel more gloomy.

7 Conclusions and Future Work

7.1 Conclusions

In this project, we built an end-to-end framework for transformer-based emotion-controlled text rewriting. The system combines a BART-based rewriter with an automatic evaluation pipeline that includes a BERT emotion classifier and Sentence-BERT semantic similarity scoring.

Our experiments show that the classifier performs strongly on the filtered GoEmotions dataset, achieving about 91% accuracy and a macro-F1 score around 0.78. The rewriting model can generate outputs that reflect different target emotions, and qualitative examples demonstrate that it can preserve the core meaning of the source sentence to a reasonable extent. At the same time, both qualitative and quantitative results confirm that balancing emotional expressiveness and semantic preservation remains the main challenge of the task.

7.2 Future Work

There are several directions for future improvement. First, the pseudo-parallel rewrite dataset could be improved to provide stronger supervision for the BART model. Second, the rewriting model could be further optimized to improve fluency, emotional precision, and semantic stability. Third, the evaluation framework could be expanded with additional semantic and stylistic metrics. Finally, future work could analyze the trade-off more systematically by plotting and studying a Pareto-style relationship between emotional intensity and semantic loss.

8 References

References

- Alok Debnath, Yvette Graham, and Owen Conlan. 2024. [Emo-gen BART - a multitask emotion-informed dialogue generation framework](#). In *Proceedings of the 1st Workshop on Simulating Conversational Intelligence in Chat (SCI-CHAT 2024)*, pages 70–74, St. Julians, Malta. Association for Computational Linguistics.
- Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. 2020. [GoEmotions: A dataset of fine-grained emotions](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4040–4054, Online. Association for Computational Linguistics.
- Zhiting Hu, Zichao Yang, Xiaodan Liang, Ruslan Salakhutdinov, and Eric P. Xing. 2017. [Toward controlled generation of text](#). In *Proceedings of the 34th International Conference on Machine Learning*, pages 1587–1596. PMLR.
- Nitish Shirish Keskar, Bryan McCann, Lav R. Varshney, Caiming Xiong, and Richard Socher. 2019. [CTRL: A conditional transformer language model for controllable generation](#).
- Mike Lewis, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. [BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 7871–7880. Association for Computational Linguistics.
- Tianxiao Shen, Tao Lei, Regina Barzilay, and Tommi Jaakkola. 2017. [Style transfer from non-parallel text by cross-alignment](#). In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Enrica Troiano, Roman Klinger, and Sebastian Padó. 2020. [Lost in back-translation: Emotion preservation in neural machine translation](#). In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 4340–4354, Barcelona, Spain (Online). International Committee on Computational Linguistics.